Text Box 2: *Rejuvenated vs relict landscapes*

The concept of rejuvenated landscape is detailed in the Depicker et al. (2020a). This is a geomorphological parameter that is commonly not used in landslide susceptibility assessment (Reichenbach et al., 2018). One of the novelties of the research carried out by Depicker et al. (2020a) is to highlight the importance of considering such a geomorphological context when studying the landslide distribution in tectonically active landscapes. In short, the four sub-catchments that we study here are situated within a mountain range – namely the North Tanganyika - Kivu Rift region in the western branch of the East African Rift - that has been developing through rifting over the last ca. 11 Ma. The associated tectonic uplift has created a specific structure where the center of the mountain range has "collapsed". This collapsed part is where Lake Kivu is located. In terms of geomorphological context, the hillslopes draining towards Lake Kivu are younger that the hillslopes draining off the mountain range. The younger hillslopes are called "rejuvenated landscape" and the older hillslopes "relict landscapes". The difference of topographic age is highly associated with the incision of the rivers in the landscapes and the presence of knickpoints; i.e. dynamic convex oversteepening of the river longitudinal profiles (a waterfall being an knickpoint). These hillslope age and incision differences have an implication on, e.g. the weathering of the rock, and hence the availability of regolith material (i.e. the slope material where most shallow landslides are usually occurring). In addition, the mountain morphology is associated with climatic conditions that are different whether we are in the rejuvenated or in the relict landscapes. The Kamembe-Gihundwe sub-catchment is located in the rejuvenated landscape, while the other three are in the relict landscape.

2.4.3 Results

Note: for further technical details on the application of the logistic regression and all the other statistical analysis performed for the landslide susceptibility assessment we encourage the reader to look at the paper published by Depicker et al (2020b) for the lake Kivu region as we follow their approach. Below are the main outcomes of our analysis in Rwanda.

2.4.3.1 Calibration and validation of the landslide susceptibility model

We used logistic regression to predict the presence/absence (1/0) of landslides. Logistic regression is an excellent data-driven multivariate modelling tool to predict binary events (Hosmer and Lemeshow, 2000), and is applied more than any other technique in the context of landslide susceptibility modelling (Reichenbach et al., 2018). The dependent variable will take values in a continuous range between 0 and 1.